The Hillingdon Hospitals
NHS Foundation Trust

Maternity Early Warning Scoring (MEWS) - Early Recognition of the Severely Ill Pregnant Woman

Author name and job title	Shabana Syed, Consultant Obstetrician Gillian Pearce, Maternity Governance Midwife
Guideline Lead name & job title	Shabana Syed, Consultant Obstetrician
Guideline Number	665
Version Number	3.0
Date Issued	26th March 2020
Next Version Date	19th March 2025
Approval Group	Maternity Guidelines Group
Approval Date	11th February 2020
Ratified Group	Clinical Outcomes and Effectiveness Group (Chair)
Ratified Date	19th March 2020

Change History

Version	Author	Summary of changes
2.0	S Syed Locum Consultant G Pearce Maternity Governance Midwife	Reformatted and full review of guideline
3.0	G Pearce Maternity Governance Midwife	Full review

Uncontrolled once printed.
It is your responsibility to check against the intranet that this printout is the most recent version of this document.

Contents

1	Aims / Purpose
2	Introduction
3	The Guidance
3.1	Process for the use of a maternity early warning scoring system (MEWS Chart)
3.2	Responsibilities of relevant staff groups
3.3	Referral to other clinicians
3.4	Indications for high dependency care
3.5	Maternity service's expectations in relation to staff training, as identified in the training needs analysis, regarding the recognition of severely ill women
3.6	Maternity service's expectations in relation to staff training, as identified in the training needs analysis, regarding maternal resuscitation
3.7	Risk Management
4	Associated Guidelines, Polices and Documents
5	Training need identified
6	Implementation and Monitoring
7	References
10	Appendices

1.0 Aim / Purpose

The scope of this guideline is to:-

- highlight clear pathways for early recognition of severely ill patients,
- highlight clear communication between senior staff
- clear communication between different specialties using the mnemonic Situation, Background, Assessment, Recommendation (SBAR)
- Responsibilities of different staff groups
- Role of maternity early warning scores (MEWS) chart
- establish the best place and care pathway for such patients

2.0 Introduction

The confidential enquiries have consistently shown that there remain a significant number of maternal deaths associated with suboptimal care. The detection of severe illness in mothers can be challenging due to a combination of the relative rarity of such events and the physiological changes associated with pregnancy. Increased physiological reserves may conceal the development of serious pathology.

3.0 The Guidance.

3.1 Process for the use of a Maternity Early warning Scoring system (MEWS Chart) – See Appendix 1

- All women admitted to the maternity unit must have their observations charted on the MEWS chart. (If a woman is having a blood transfusion, the observations are recorded on the MEWS / HDU chart. If the woman is admitted in labour to the Midwifery Led Unit, observations are documented on the partogram. The blood transfusion chart should include the number and labels of blood products and state on the observation side “to refer to MEWS chart.”)

- Women in labour will have their observations documented on the partogram unless clinically indicated.

- Early recognition can be aided with the use of the Maternity Early Warning Scores (MEWS) chart.

- Their systematic use will trigger a response to identify women in whom more frequent and detailed observations and examination are required.

- The need for a response to abnormal observations as detailed in the trigger flow chart (**Appendix 1**- 2nd part) enables healthcare professionals to escalate care early.

- The measurement of observations and how they score on the chart will give a final score which may trigger a response for review by a doctor.

Physiological observations

In order to obtain a MEWS score, a complete set of physiological observations are required which include all of the following (see **Appendix 2**);

- Respiratory rate
- Pulse rate
- Blood pressure

- Temperature
- Oxygen saturation levels (SpO2)
- Conscious level
- Urine output

The frequency of observations on the antenatal and postnatal ward will depend on the clinical risk assessment and individualised management plans of care documented in the health records. See Recovery Guideline for the frequency of observations following operative procedures. High risk women will have their observations done 4 hourly unless otherwise clinically indicated. Low risk women post spontaneous vaginal delivery will have their observations done post-delivery, on arrival to the postnatal ward and then daily whilst in hospital.

Respiratory rate:
Respiratory rate is the most sensitive indicator of deteriorating physiology and must be recorded in all women every time a full set of observations are taken.

Respiratory rate is the best marker of a sick woman and is the first observation that will indicate a problem or deterioration in condition. Respiratory rate is a mandatory observation

Pulse rate:
Tachycardia is a highly significant sign of an unwell woman. Pulse rate can be monitored via a saturation probe on the finger. Cautions regarding Pulse rate:-

If the woman is peripherally shut down in cases of haemorrhage
- The pulse oximetry probe will not detect the pulse accurately.
- Pulse properties such as volume and regularity cannot be assessed
- Nail varnish affects wave form accuracy

A persistent sinus tachycardia must be referred to an obstetric registrar or consultant obstetrician.

Oxygen Saturation:
Oxygen saturations are not routinely monitored on pregnant women unless they become unwell, or have an increase or decrease in their respiratory rate or medical/obstetric condition necessitates use.

However, all women who TRIGGER require their oxygen saturations to be monitored: -
- Normal > 95% on air
- If on O2 therapy record percentage of O2 in use [%]
- If saturation monitoring is not required, in the clinical judgement of the midwife, record ‘not applicable’ on the MEWS chart.

Blood pressure:
Use of the correct cuff size for the woman is vitally important for the accuracy of recordings of blood pressure especially in the obese woman.

All pregnant women with a systolic blood pressure of 140mm/Hg / 90 mm/Hg or more require anti -hypertensive treatment and a reading of 160 triggers a red score.

Falling BP should be regarded as a late sign of deterioration [perimortem] as pregnant women can lose up to 30-40% of their circulating blood volume with no change to their vital signs especially BP.

Caution: Electronic recordings of blood pressure can underestimate readings by up to 5%.

It is recommended good practice that if blood pressure is raised on dynamat readings this should be rechecked manually at least once using an aneroid BP machine.

Urine output:

- Urine output must be charted on a separate fluid balance chart.
- In addition the hourly total must be charted on the MEWS chart
- For Post operative women and unwell women, use HDU fluid balance chart
- The optimum urine output is 1ml/kg/hr, the minimum urine output 0.5ml/kg/hr

Additional Scores:
These include sedation score, pain score and nausea and vomiting score.

Recognition of deterioration using MEWS chart

- If a patient triggers one red or two yellow scores at any one time the obstetric specialty doctor and/or anaesthetist must be asked to review the patient, start treatment and make a management plan.
- Small changes in monitoring in comparison to earlier recordings may indicate early deterioration of a patient, before the onset of obvious changes in an individual.
- Of the parameters measured, respiratory rate is considered to be the most sensitive indicator of a patient's physiological wellbeing and any rises must trigger a review by an obstetric specialty doctor or anaesthetic doctor.
- If the patient appears agitated or confused this must trigger an obstetric specialty or anaesthetic doctor review

Recognition of deterioration in addition to MEWS chart

The physiological parameters recorded on the MEWS chart are not all inclusive. Some other observations performed by medical practitioners are important to the identification of deterioration in the woman's condition include:

- Symphisis fundal height; it is important to note the height of the fundus. Any rise could indicate haemorrhage, which may be concealed.
- Increase in abdominal girth or signs of peritonism.
- Capillary refill which should be less than 3 seconds, delay indicates hypovolaemia
- Temperature of peripheries
- Hemacue testing: Hemacue machines are available in all clinical areas and can be used in all cases of haemorrhage to obtain immediate haemoglobin estimation.
- Assessment of agitation
- Heart disease can occur for the first time in pregnancy. Severe chest pain must be escalated to obstetric registrar or consultant obstetrician.

3.2 Responsibilities of relevant staff groups

Role of the Maternity Support Worker (MSW) /Midwife/Theatre Nurse:-
The MEWS chart must be filled in contemporaneously

Triggers on the MEWS chart acted upon immediately by request for senior review by an obstetric specialty doctor or anaesthetist.

They must ensure all care is documented in the maternity record.

MSWs charting abnormal findings must get the midwife to recheck them and review the woman.

If the problem persists after initial treatment the senior obstetric specialty doctor/ consultant must be informed

Role of the Labour Ward coordinator:-
Will ensure one to one care is provided for any unstable or high risk patient.

Concerns will be communicated immediately to the obstetric specialty doctor or consultant. They must be present at the labour ward round reviewing all patients.

They must ensure timely review by the obstetric specialty doctor or anaesthetist if required and escalate to the consultant, if this review does not take place.

Obstetric Team:-
Must attend promptly when informed of a seriously ill woman or trigger on the MEWS chart If unable to attend they must escalate to the anaesthetic team or obstetric consultant.

Will review at a minimum 4hrly and attend promptly when asked to review.

Must inform the obstetric consultant and ITU specialists of rapid deterioration

Anaesthetic team:-
They must be involved early in the assessment and care of all seriously ill women with a multidisciplinary assessment and review.

Can be called instead of the obstetric team if the obstetric team is unavailable for initial review.

There must be a multidisciplinary round at least once per day.

They must escalate concerns to the consultant and other relevant clinicians.

They must document their management of care at each review.

The anaesthetic consultant must be informed of all management plans.

Obstetric Consultant: Must attend for:-

- Eclampsia
- Maternal collapse (such as massive abruption, septic shock)
- Caesarean section for major placenta praevia
- Postpartum haemorrhage of more than 1.5litres where the haemorrhage is continuing and massive obstetric haemorrhage protocol has been started.
- Return to theatre – laparotomy
- Uterine rupture
- Seriously ill women who are unstable

- When requested

The consultant obstetrician remains the lead clinician. There must be communication at a senior level between consultants which must be clearly documented.

Obstetric medicine consultants are available for advice via switchboard if needed.

> If a patient deteriorates rapidly, the obstetric consultant must be informed immediately, and the anaesthetic team must be notified before referring to other specialties.
> If a deteriorating patient cannot be reviewed within 15 mins by a doctor if requested, the anaesthetic and obstetric consultant must be informed

3.3 Referral to other clinicians outside the maternity service using SBAR

1. Within the maternity unit:

At recognition and diagnosis of severe illness, after review by the obstetric team, any patients with any system failing e.g. Cardiovascular – hypotension in sepsis, must be discussed with:-

- Labour Ward coordinator
- Obstetric consultant
- Anaesthetic consultant

2. Outside the maternity unit. COMMUNICATION at a SENIOR LEVEL IS ESSENTIAL as other specialists may be unfamiliar with pregnancy physiology

- Relevant specialty according to pre-existing illness
- Relevant specialty according to current working diagnosis (see table below for guidance which is not exhaustive). If in doubt discuss with consultant obstetrician.
- ITU/HDU staff

Diagnosis	Specialty to refer
Sepsis	Microbiology/ITU/Medicine
Respiratory	Microbiology/ITU/Medicine/Respiratory
Haemorrhage/DIC	Haematology/ITU/Medicine
Cardiac	Medicine/ITU/Cardiology
VTE/Amniotic fluid embolism	Haematology/ITU/Medicine
Neurological	ITU/Medicine/Neurology
Surgical	Surgeons/ITU

When the pregnant woman has a pre-existing co-morbidity under the care of another specialist, they must be informed with close communication at a senior level.

Intensive care specialists must be informed of seriously ill women or anyone who has rapidly deteriorated. They will review them on the labour ward especially if ITU transfer is being considered.

3.4 Indications for high dependency care

This must be considered if the woman remains unstable and discussed at a senior level. The outcome of this discussion must be documented in the maternity record.

- The decision to manage a patient in high dependency care rests with a senior obstetrician and/ or anaesthetist.

- HDU care can be administered on the labour ward or in the general hospital (see High Dependency Care Guideline)

- There are a significant number of women with a high acuity of illness necessitating a level of cardiovascular and respiratory monitoring that exceeds 'normal' practice in delivery units.

- It has been estimated that 5% of women require high dependency care such as in the indication table below.

Indication The following list is not exhaustive, and each case must be considered on an individual basis.
Women who require hourly monitoring
Severe Pregnancy- induced hypertension and pre-eclampsia, including HELLP
Eclampsia
Severe antepartum and postpartum haemorrhage (>1500ml)
Severe sepsis
Sudden unexplained collapse
DIC
Uncontrolled diabetes
Certain cardiac, pulmonary and neurological conditions
Elevated MEWS score

If the woman continues to deteriorate:

The woman may need transfer to ITU or HDU care in the general hospital-

- Bleep 2222 and state patient name, hospital number and lead consultant.
- Contact the Outreach team for assistance (Bleep 5230) and for assessment for transfer to HDU (see HDU guideline)
- If out of hours contact clinical site practitioners (Bleep 5838)
- Contact Medical/Surgical SPR depending on clinical picture via switchboard
- A Consultant to Consultant referral is required if Medical/Surgical Consultant opinion sought.

Management

- Ensure a diagnosis is sought if not yet reached
- Ensure there is a clear documented individual management plan at each review
- Treatment instituted promptly

- Investigations are done promptly
- Required Level of monitoring is reviewed by senior doctors (ST5 or above)
- Regular review by senior doctors (ST5 or above)
- Multidisciplinary communication occurs and is documented.
- Woman and relatives are debriefed

If the woman becomes unresponsive or pulseless or deteriorates significantly, contact the maternity emergency team on 2222. Initiate IMMEDIATE maternal resuscitation.

3.5 Maternity service's expectations in relation to staff training, as identified in the training needs analysis, regarding the recognition of severely ill women
See section 5.

3.6 Maternity service's expectations in relation to staff training, as identified in the training needs analysis, regarding maternal resuscitation
See section 5.

3.7 Risk Management

All severely ill women must be reported on DATIX (incident form) and to relevant external organisation e.g. UKOSS or HPA.

4 Associated Guidelines, Polices and Documents

- Recovery and Post Anaesthetic Care Guideline
- Emergency Care for Pregnant Patients Presenting Outside Maternity Guideline
- Sepsis in Maternity Guideline
- Antenatal Management of Hypertension in Pregnancy and Pre-eclampsia
- Postnatal Management of Hypertension and Pre-Eclampsia
- Severe Pre-eclampsia and Eclampsia
- Maternity Specific Mandatory Training Policy, Including Skills and Drills

5 Training need identified

- Any specific training needs will be identified by the clinical line manager, clinical lead and Practice Development Team; Training will be discussed and implemented.
- Annual training on recognition of the severely ill pregnant woman will take place as part of professional training days for the multidisciplinary team as specified in the Training Needs Analysis (TNA).
- All clinical staff working with patients should have training in maternal resuscitation which is covered in the skills and drills days.

6 Implementation and Monitoring

All guidelines, protocols and procedures are available on the intranet. All staff have a responsibility to check the Trust's intranet and ensure that they are referring to the most recent issue of the document. The weekly general information bulletin includes notification of publication of new/updated documents.
This clinical guideline, protocol or procedure will be monitored using local and National guidance and via the Trust Incident and Risk Management Policies. If risks/issues are

identified further specific monitoring will be considered, for example, clinical audit or risk assessment.

7 References

1. MBRRACE –UK (2019) Mothers and Babies Reducing Risk through Audits and confidential enquiries
2. Royal College of Obstetricians and Gynaecologists. (2019). *Maternal Collapse In Pregnancy And The Puerperium.* London. RCOG. Available at: www.rcog.org.uk
3. PROMPT rcog press 2012
4. MOET handbook

8 Appendices

- Appendix 1 - MEWS chart
- Appendix 2 - Frequency of MEWS observations
- Appendix 3 - SBAR - escalation

Appendix 1

Hospital No.

Surname

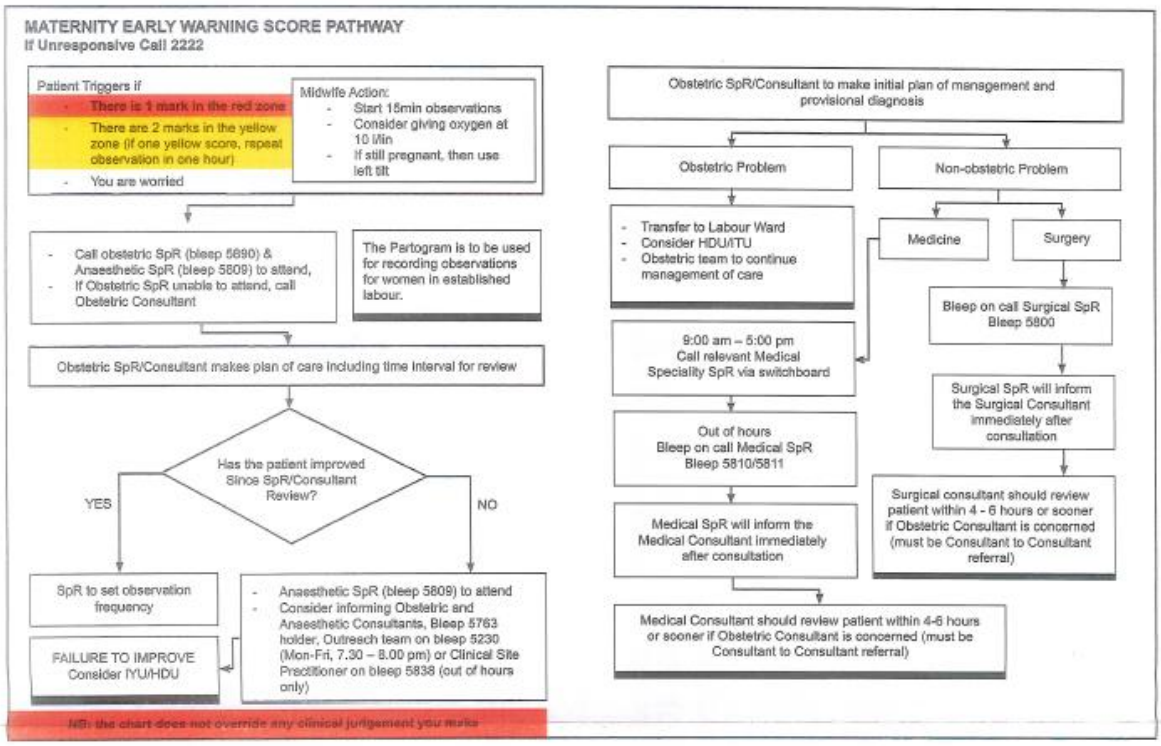
34167 12/16

MATERNITY EARLY WARNING SCORE (MEWS) CHART

The Hillingdon Hospitals NHS
NHS Foundation Trust

Ward:

| | | |



The Hillingdon Hospitals NHS Foundation Trust

Actions Taken if 2 or more Yellow / 1 more of Red Observations		
Observation Triggered Date and Time	Action taken	Signature and Print Name

Appendix 2

Frequency of MEWS observations

Caesarean sections and other procedures under anaesthesia

For the first 30mins

MONITOR	FREQUENCY
Heart rate Blood pressure Oxygen saturation	Record every 5 minutes
Respiratory rate	Record every 15 minutes If <10/min or >20/min increase frequency
Temperature	On admission If outside normal parameters (36-37.2), repeat in 30 mins

Once stable, after 30 minutes record above observations (except temperature) half hourly for 2 hours, then hourly thereafter until discharge criteria met (See Recovery and post anaesthetic care guideline)

Postpartum haemorrhage

- ½ hourly for four hours (remain on CDS if possible)
- Four hourly for 24 hours
- Prophylactic syntocinon Four hourly for 24 hours

Hypertension/Pre-eclampsia

- **NB Women with diastolic of 110 or over and/or MAP of 125 or above should commence on the Pre-eclampsia protocol**
- Minimum four hourly until discharge or start of labour

Identified antenatal co-morbidities in patients (diabetics, ruptured membranes, known or suspected infection)

- Four hourly (unless indicated otherwise) until discharge

Postnatal women with suspected or confirmed infection

- Four hourly for a minimum of 24 hours
- Once daily until discharge

Medical management of miscarriage

- Hourly from start of treatment until delivery

Blood transfusion

- Prior to start of transfusion
- 15 minutes into transfusion
- Post transfusion

(Must be done for each unit of blood transfused)

This is a minimum requirement for observation frequency and midwives and medical staff should use their clinical judgement in each individual case.

Reasons to trigger action checklist

- EWS ≥ 3
- Patient causing concern
- Diastolic BP ≥ 110 mm/Hg
- SpO2 <90% on air or <94% on 5 litres O2

Appendix 3

SBAR FOR ESCALATION

SBAR given by: (Print and sign) _____
SBAR escalated to: (Name of person called and designation)_____
Date: _____ Time: _____

S

- **Situation**
  - Your name
  - Designation
  - Ward
  - Patient 's name
  - I am calling because I am concerned that

B

- **Background**
  - Admission date:
  - Reason of admission
  - Parity
  - Gestation
  - Singleton/Multiple pregnancy
  - Past medical history and medications
  - Investigations and results; e.g. relevant USS
  - Diagnosis

A

- **Assessment**
  - Observations
  - MEWS score
  - Pain score (if applicable)
  - Urine output (if applicable)
  - PV loss/liquor/lochia/uterus/EBL (if applicable)
  - Clinical impression

R

- **Recommendation**
  - I need you to .....
  - Come to see the patient in the next (XX) minutes
  - AND
  - Is there anything I need to do in the meantime?